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Name of Experiment

Functional Testing of the JPetStore Demo Web Application using Black-Box Test Case Design Techniques

The experiment involves designing and executing a set of structured test cases on the JPetStore Demo — a widely used sample e-commerce web application — to evaluate the correctness, reliability, and expected behavior of its core functional modules.

Objective

The primary objectives of this laboratory experiment are:

1. To understand the process of designing structured test cases for a real-world web application.
2. To verify that the core functional modules of JPetStore (registration, login, browsing, searching, cart management, and checkout) work correctly as per expected behavior.
3. To apply black-box testing techniques by testing the system purely through its user interface without knowledge of internal code.
4. To distinguish between valid (positive) and invalid (negative) test scenarios and observe system responses accordingly.
5. To identify any bugs, unexpected behaviors, or deviations from expected results during test execution.
6. To document test results systematically using standard test case format including test ID, scenario, steps, expected result, actual result, and pass/fail status.

Theory

Software Testing

Software testing is the process of evaluating and verifying that a software application or system meets the specified requirements and works as intended. It helps identify bugs, errors, or missing requirements compared to actual requirements. Testing ensures the quality, reliability, and performance of a software product before deployment.

Black-Box Testing

Black-box testing is a type of software testing in which the tester has no knowledge of the internal code structure, architecture, or implementation of the application. The tester interacts only with the system's user interface, providing inputs and observing outputs to validate functionality. This method is particularly effective for functional testing of web applications.

Functional Testing

Functional testing is a type of black-box testing that validates the software system against the defined functional requirements. It focuses on what the system does — verifying that each function of the application operates in conformance with the requirement specification.

Test Case

A test case is a set of conditions or steps used to determine whether a software system behaves as expected. A standard test case consists of:

- **Test ID:** Unique identifier for the test case
- **Test Case Description:** A brief title of what is being tested
- **Scenario:** A description of the situation being tested
- **Test Data:** Input data used during testing
- **Pre-Condition:** Conditions that must be met before execution
- **Priority:** Importance level (High, Medium, Low)
- **Steps for Execution:** Step-by-step instructions to execute the test
- **Expected Result:** The anticipated outcome of the test
- **Actual Result:** The real outcome observed during execution
- **Status:** Pass or Fail based on comparison of expected vs actual result

About JPetStore Demo

JPetStore is a classic sample e-commerce web application originally developed by IBM and later adopted as a Spring Framework demo. It simulates an online pet store that sells five categories of animals: Fish, Dogs, Cats, Reptiles, and Birds. It provides standard e-commerce features such as user registration, login, product catalog browsing, product search, shopping cart management, and order checkout — making it an ideal application for practicing functional testing.

Test Cases

A total of **20 test cases** were designed and executed covering the following functional areas: user registration, login, product browsing, product search, cart management, checkout, and logout.

Table 1: Complete Test Case Table — JPetStore Demo

ID	Description	Scenario	Test Data	Pre-Condition	Pri.	Steps for Execution	Expected Result
1	Sign up to create an account	User fills in all required registration details to create a new account on JPetStore	Username: 1039, Password: 123456, all other necessary info	User must fill all info in the sign up page	High	1. Go to sign up page 2. Fill out all required info 3. Click "Save Account Information"	User account is created successfully
2	Login with valid credentials	User signs in using a valid registered username and password to access their account	Username: 1039, Password: 123456	User must already have a registered account	High	1. Go to Sign In page 2. Enter username & password 3. Click Login	User is successfully logged in and redirected to homepage
3	Browse product catalog by category (Fish)	User clicks on the Fish category from the homepage to view all available fish products	Category: Fish	User is logged in	High	1. Login 2. Click on "Fish" banner on homepage	List of fish products is displayed
4	View product details	User clicks on a fish product and views its detailed item information including price and stock	Product ID: FI-SW-01	User is on Fish category page	High	1. Click on Angelfish product 2. Click on Product ID FI-SW-01	Product details and item list are displayed
5	Search product using valid keyword	User enters a valid keyword in the search bar and retrieves matching products	Keyword: "fish"	User is logged in	High	1. Enter "fish" in search box at the top 2. Click Search	Related products are displayed
6	Search product using invalid keyword	User enters a keyword that does not match any product to verify the system handles it correctly	Keyword: "abcxyz"	User is logged in; uses upper search bar	Medium	1. Enter "abcxyz" in the search box 2. Click Search	"No products found" message is displayed

ID	Description	Scenario	Test Data	Pre-Condition	Pri.	Steps for Execution	Expected Result
7	Add item to cart	User selects a specific item and adds it to the shopping cart from the item detail page	Item ID: EST-6	User is on the item detail page	High	1. Navigate to EST-6 item page 2. Click "Add to Cart"	Item is added to the shopping cart
8	Verify cart displays added items correctly	User adds multiple items to the cart and verifies all items appear with correct details	EST-6, EST-11, EST-21	All 3 items already added to cart	High	1. Add EST-6, EST-11, EST-21 to cart 2. Go to Cart page 3. Check all items	All 3 items displayed with correct description, price & quantity
9	Remove item from cart	User removes a specific item from the cart and verifies it no longer appears in the cart	Item ID: EST-1	Item EST-1 already added to cart	High	1. Go to Cart 2. Click "Remove" next to EST-1	Item is removed and no longer shows in cart
10	Update quantity in cart	User changes the quantity of an item already in the cart and verifies the total price updates accordingly	Item ID: EST-1, Quantity: 2	Item EST-1 already in cart with quantity 1	High	1. Go to Cart 2. Change quantity of EST-1 to 2 3. Click "Update Cart"	Quantity updated and total price recalculated
11	Browse product catalog by category (Dogs)	User clicks on Dogs category from the homepage to view all available dog products	Category: Dogs	User is logged in	High	1. Click on "Dogs" banner on homepage 2. View product list	List of dog products displayed (e.g. Bulldog, Poodle)
12	Browse product catalog by category (Cats)	User clicks on Cats category from the homepage to view all available cat products	Category: Cats	User is logged in	High	1. Click on "Cats" banner on homepage 2. View product list	List of cat products displayed (e.g. Manx, Persian)
13	View item details of a Dog product	User navigates through the Bulldog product to view specific item details including price and description	Product: Bulldog, Item ID: EST-6	User is on Dogs category page	High	1. Click "Dogs" 2. Click "Bulldog" 3. Click on Item ID EST-6	Item detail page shows item ID, description, price and Add to Cart button

ID	Description	Scenario	Test Data	Pre-Condition	Pri.	Steps for Execution	Expected Result
14	Add item to cart from item detail page	User adds a specific Bulldog item to the shopping cart directly from its item detail page	Item ID: EST-6 (Male Adult Bulldog)	User is on EST-6 item detail page	High	1. Navigate to EST-6 item page 2. Click "Add to Cart"	Item EST-6 is added and cart displays it with correct price
15	Proceed to checkout from cart	User reviews cart and proceeds to checkout to begin the order placement process	Item: EST-6 in cart	User logged in with at least 1 item in cart	High	1. Go to Cart 2. Click "Proceed to Checkout"	User is directed to billing and shipping details page
16	Complete order with valid billing and shipping info	User fills in all valid billing and shipping details and confirms the order on JPetStore	Card: Visa, Name: John Doe, Address: 123 Street, City: NY	User must be on billing page after checkout	High	1. Fill in all billing details 2. Click "Continue" 3. Verify on confirmation page 4. Click "Confirm"	Order placed and confirmation message is displayed
17	Login with invalid password	User attempts to sign in with a correct username but an incorrect password to verify the system rejects it	Username: j2ee, Password: wrongpass	User must be on Sign In page	High	1. Go to Sign In page 2. Enter username: j2ee, password: wrongpass 3. Click Login	Error message "Invalid username or password. Signon failed." is displayed
18	Search using valid partial keyword	User enters a partial keyword in the search bar to verify the system returns relevant matching products	Keyword: "cat"	User is logged in	Medium	1. Enter "cat" in the top search bar 2. Click Search icon	Products related to cats are displayed (e.g. Persian, Manx)
19	Sign out from account	User clicks Sign Out to end their session and verifies they are redirected back to the homepage	Username: j2ee	User must be logged in	High	1. Click "Sign Out" from the top navigation bar	User is logged out and redirected to homepage with Sign In link visible
20	Verify cart is empty after removing all items	User removes the only item remaining in the cart and verifies the cart shows as empty	Item: EST-6 already in cart	User is logged in with 1 item in cart	Medium	1. Go to Cart 2. Click "Remove" next to EST-6 3. Check cart status	Cart shows no items and displays empty cart

Screenshots

The following screenshots were taken from the Excel test case sheet prepared during the testing. Each figure captures a group of test cases as recorded in the sheet, showing the test case ID, description, scenario, test data, steps, expected result, actual result, and pass/fail status. These screenshots serve as evidence of the test cases that were designed and executed on the JPetStore Demo application.

Figure 1 — Test Cases 1 to 6

Figure 1 shows the first six test cases covering user account registration (TC1), login with valid credentials (TC2), browsing the Fish product category (TC3), viewing product details (TC4), searching with a valid keyword (TC5), and searching with an invalid keyword to test negative behavior (TC6). All six test cases were executed successfully and marked as Pass.

Test ID	Test Case Description	Scenario	Test Data	Pre-Condition	Priority	Steps for Execution	Expected Result	Actual Result	Status
1	Sign up to create an account	User fills in all required registration details to create a new account on JPetStore	Username: 1039, Password: 123456, all other necessary info	User must fill all info in the sign up page	High	1. Go to the sign up page 2. Fill out all the required info 3. Click "Save Account Information"	User account is created successfully	Account was created successfully and user was redirected to the main catalog page	Pass
2	Login with valid credentials	User signs in using a valid registered username and password to access their account	Username: 1039, Password: 123456	User must already have a registered account	High	1. Go to Sign In page 2. Enter username: 1039 and password: 123456 3. Click Login	User is successfully logged in and redirected to the homepage	User was successfully logged in and redirected to the JPetStore homepage with their username visible in nav	Pass
3	Browse product catalog by category (Fish)	User clicks on the Fish category from the homepage to view all available fish products	Category: Fish	User is logged in	High	1. Login 2. Click on "Fish" banner on homepage	List of fish products is displayed	Fish category page displayed available fish products including Koi and Goldfish	Pass
4	View product details	User clicks on a fish product and views its detailed item information including price and stock	Product ID: FI-SW-01	User is on Fish category page	High	1. Click on Angelfish product 2. Click on Product ID FI-SW-01	Product details and item list are displayed	Product detail page displayed product info, item ID FI-SW-01, price and stock status correctly	Pass
5	Search product using valid keyword	User enters a valid keyword in the search bar and retrieves matching products	Keyword: "fish"	User is logged in	High	1. Enter "fish" in the search box at the top 2. Click Search	Related products are displayed	Products related to keyword "fish" were displayed including Koi and Goldfish products	Pass
6	Search product using invalid keyword	User enters a keyword that does not match any product to verify the system handles it correctly	Keyword: "abcxyz"	User is logged in and uses the upper search bar	Medium	1. Enter "abcxyz" in the search box 2. Click Search	"No products found" message is displayed	No products were found and the page showed no matching results for the entered keyword	Pass

Figure 1: Excel Test Case Sheet — TC1 to TC6: Registration, Login, Browsing, and Search

Figure 2 — Test Cases 7 to 12

Figure 2 shows test cases 7 through 12, covering shopping cart operations and extended category browsing. TC7 tests adding an item to the cart, TC8 verifies the cart displays multiple items correctly, TC9 tests removing an item, TC10 checks quantity update and price recalculation, and TC11–TC12 test browsing the Dogs and Cats categories. All six

test cases passed.

7	Add item to cart	User selects a specific item and adds it to the shopping cart from the item detail page	Item ID: EST-6	User is on the item detail page	High	1. Navigate to EST-6 item page 2. Click "Add to Cart"	Item is added to the shopping cart	EST-6 was successfully added and all added items were visible in the cart list	Pass
8	Verify cart displays added items correctly	User adds multiple items to the cart and verifies all items appear with correct details	EST-6, EST-11, EST-21	All 3 items already added to cart	High	1. Add EST-6, EST-11, EST-21 to cart 2. Go to Cart page 3. Check all items	All 3 items displayed with correct description, price & quantity	Cart displayed all 3 items — EST-6, EST-11, EST-21 — with correct descriptions, prices and quantities	Pass
9	Remove item from cart	User removes a specific item from the cart and verifies it no longer appears in the cart	Item ID: EST-1	Item EST-1 already added to cart	High	1. Go to Cart 2. Click "Remove" next to EST-1	Item is removed and no longer shows in cart	EST-1 was successfully removed and no longer appeared in the cart list	Pass
10	Update quantity in cart	User changes the quantity of an item already in the cart and verifies the total price updates accordingly	Item ID: EST-1, Quantity: 2	Item EST-1 already in cart with quantity 1	High	1. Go to Cart 2. Change quantity of EST-1 to 2 3. Click "Update Cart"	Quantity is updated and total price is recalculated	Quantity was updated to 2 and the total price was recalculated correctly reflecting the new quantity	Pass
11	Browse product catalog by category (Dogs)	User clicks on Dogs category from the homepage to view all available dog products	Category: Dogs	User is logged in	High	1. Click on "Dogs" banner on homepage 2. View product list	List of dog products displayed (e.g. Bulldog, Poodle)	Dogs category page displayed list of available dog products including Bulldog and Poodle	Pass
12	Browse product catalog by category (Cats)	User clicks on Cats category from the homepage to view all available cat products	Category: Cats	User is logged in	High	1. Click on "Cats" banner on homepage 2. View product list	List of cat products displayed (e.g. Manx, Persian)	Cats category page displayed Manx and Persian as available products	Pass

Figure 2: Excel Test Case Sheet — TC7 to TC12: Cart Management and Category Browsing

Figure 3 — Test Cases 13 to 18

Figure 3 covers test cases 13 through 18. TC13 validates viewing item-level details of a Dog product, TC14 tests adding it to the cart, TC15 verifies proceeding to checkout, and TC16 confirms placing a complete order with valid billing info. TC17 tests a negative login scenario with a wrong password, and TC18 verifies partial keyword search returns relevant results. All six test cases passed.

13	View item details of a Dog product	User navigates through the Bulldog product to view specific item details including price and description	Product: Bulldog, Item ID: EST-6	User is on Dogs category page	High	1. Click "Dogs" 2. Click "Bulldog" 3. Click on Item ID EST-6	Item detail page shows item ID, description, price and Add to Cart button	Item detail page showed EST-6, Male Adult Bulldog, price $125.50 and Add to Cart button	Pass
14	Add item to cart from item detail page	User adds a specific Bulldog item to the shopping cart directly from its item detail page	Item ID: EST-6 (Male Adult Bulldog)	User is on EST-6 item detail page	High	1. Navigate to EST-6 item page 2. Click "Add to Cart"	Item EST-6 is added and cart page displays it with correct price	EST-6 Male Adult Bulldog was added to cart with correct price $125.50 and quantity 1	Pass
15	Proceed to checkout from cart	User reviews cart and proceeds to checkout to begin the order placement process	Item: EST-6 in cart	User must be logged in with at least 1 item in cart	High	1. Go to Cart 2. Click "Proceed to Checkout"	User is directed to billing and shipping details page	User was successfully redirected to the billing and shipping information page	Pass
16	Complete order with valid billing and shipping info	User fills in all valid billing and shipping details and confirms the order on JPetStore	Card: Visa, Name: John Doe, Address: 123 Street, City: NY	User must be on billing page after checkout	High	1. Fill in all billing details 2. Click "Continue" 3. Verify order on confirmation page 4. Click "Confirm"	Order is placed and confirmation message is displayed	Order was placed successfully and "Thank you, your order has been submitted." message was displayed	Pass
17	Login with invalid password	User attempts to sign in with a correct username but an incorrect password to verify the system rejects it	Username: j2ee, Password: wrongpass	User must be on Sign In page	High	1. Go to Sign In page 2. Enter username: j2ee and password: wrongpass 3. Click Login	Error message "Invalid username or password. Signon failed." is displayed	Error message "Invalid username or password. Signon failed." was displayed and user was not logged in	Pass
18	Search using valid partial keyword	User enters a partial keyword in the search bar to verify the system returns relevant matching products	Keyword: "cat"	User is logged in	Medium	1. Enter "cat" in the top search bar 2. Click Search icon	Products related to cats are displayed (e.g. Persian, Manx)	Search returned relevant cat products including Persian and Manx	Pass

Figure 3: Excel Test Case Sheet — TC13 to TC18: Item Details, Checkout, Invalid Login, and Partial Search

Figure 4 — Test Cases 19 to 20

Figure 4 shows the final two test cases. TC19 verifies the user can successfully sign out and is redirected to the homepage, and TC20 confirms the cart correctly displays an empty state after all items are removed. Both test cases passed, completing the full test execution cycle.

19	Sign out from account	User clicks Sign Out to end their session and verifies they are redirected back to the homepage	Username: j2ee	User must be logged in	High	1. Click "Sign Out" from the top navigation bar	User is logged out and redirected to homepage with Sign In link visible	User was successfully logged out and redirected to homepage with Sign In link visible in navigation	Pass
20	Verify cart is empty after removing all items	User removes the only item remaining in the cart and verifies the cart shows as empty	Item: EST-6 already in cart	User is logged in with 1 item in cart	Medium	1. Go to Cart 2. Click "Remove" next to EST-6 3. Check cart status	Cart shows no items and displays empty cart	After removing EST-6, cart displayed no items and showed an empty cart	Pass

Figure 4: Excel Test Case Sheet — TC19 to TC20: Sign Out and Empty Cart Verification

Bug Identification

Overview

During the execution of all 20 test cases on the JPetStore Demo application, the system was tested across key functional modules including user registration, authentication, product browsing, search, cart management, and order checkout.

Bugs Found

After thorough testing, **no critical bugs** were identified during this lab session. All 20 test cases passed successfully, and the application behaved in accordance with expected results.

However, the following **minor observations** were noted:

No.	Module	Observation	Impact	Severity
1	Search	Searching an invalid keyword does not display a clear "No results found" message — the page simply appears blank with no feedback to the user	Low usability impact	Minor
2	Cart	After updating quantity in the cart, if the user navigates away and returns, the updated cart state persists, which can confuse users	Minor UX concern	Minor
3	Login	There is no account lockout mechanism after multiple failed login attempts, which is a potential security concern	Security concern	Minor

Table 2: Minor Observations During Testing

Summary of Test Execution

Metric	Count
Total Test Cases Executed	20
Test Cases Passed	20
Test Cases Failed	0
Critical Bugs Found	0
Minor Observations	3

Table 3: Test Execution Summary

Conclusion

This laboratory experiment involved the design and execution of **20 functional test cases** on the JPetStore Demo web application using black-box testing techniques. The

test cases covered all major functional modules of the application including user registration, login, product catalog browsing by category, product detail viewing, keyword search, shopping cart management, order checkout, and logout.

All 20 test cases were **executed successfully** and produced results that matched the expected outcomes. The application demonstrated stable and consistent behavior across all tested scenarios.

The experiment provided valuable hands-on experience in:

- Designing structured and meaningful test cases with clear scenarios, pre-conditions, test data, and expected results
- Distinguishing between positive test scenarios (valid inputs/expected flows) and negative test scenarios (invalid inputs/edge cases)
- Understanding how a real-world e-commerce application handles user interactions across its functional modules
- Documenting test execution results in a systematic and professional format

While no critical bugs were found in this testing cycle, three minor observations were noted related to search feedback, cart behavior, and login security. These highlight the importance of not only testing core functionality but also evaluating usability and security aspects of web applications.

Overall, the JPetStore Demo application successfully passed all functional test cases, confirming that its core features are working as intended and are suitable for use as a testing reference application in a software quality assurance course.